

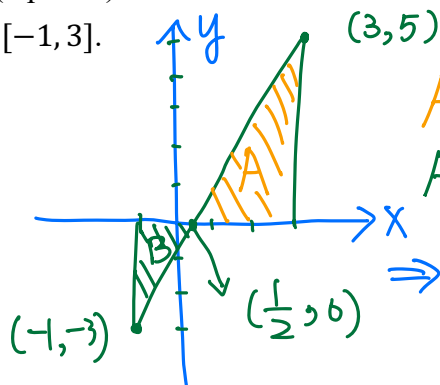
Quiz2, MAT1575 Professor Chiu

ID: _____

Name: _____

- This quiz consists of 3 questions for a total of 10 points.
- You have 15 minutes to complete the quiz.
- Show all work and justify your answers.
- Wishing you success.

(3 points) 1. Find the area between the curve of the function $f(x) = 2x - 1$ and the x-axis over the interval $[-1, 3]$.



$$\text{Area of } A = \left(3 - \frac{1}{2}\right) \cdot 5 \cdot \frac{1}{2} = \frac{25}{4}$$

$$\text{Area of } B = \left(\frac{1}{2} - (-1)\right) \cdot 3 \cdot \frac{1}{2} = \frac{9}{4}$$

$$\Rightarrow \text{Area between } f(x) = 2x - 1 \text{ and the x-axis over } [-1, 3] \text{ is } A - B = \frac{25}{4} - \frac{9}{4} = \frac{16}{4} = 4$$

(4 points) 2. Use the definition of the definite integral to evaluate $\int_{-1}^3 (2x - 1) dx$.

Hint: Use a right endpoint approximation R_n with a regular partition to generate the Riemann sum.

$$f(x) = 2x - 1, \quad I = [-1, 3], \quad \Delta x = \frac{3 - (-1)}{n} = \frac{4}{n}$$

$$P = \{-1, -1 + \Delta x, -1 + 2\Delta x, -1 + 3\Delta x, \dots, -1 + n\Delta x = 3\}$$

$$x_i = -1 + i \cdot \Delta x = -1 + \frac{4i}{n}$$

$$R_n = \sum_{i=1}^n f(x_i) \cdot \Delta x = \sum_{i=1}^n \left(2\left(-1 + \frac{4i}{n}\right) - 1\right) \frac{4}{n} = \sum_{i=1}^n \left(\frac{8i}{n} - 3\right) \frac{4}{n} = \sum_{i=1}^n \left(\frac{32i}{n^2} - \frac{12}{n}\right)$$

$$= \sum_{i=1}^n \frac{32i}{n^2} - \sum_{i=1}^n \frac{12}{n} = \frac{32}{n^2} \sum_{i=1}^n i - \frac{12}{n} \sum_{i=1}^n 1 = \frac{32}{n^2} \cdot \frac{n(n+1)}{2} - \frac{12}{n} \cdot n$$

$$= \frac{16n^2 + 16n}{n^2} - 12 = \frac{16n^2}{n^2} + \frac{16n}{n^2} - 12 = 16 + \frac{16}{n} - 12 = 4 + \frac{16}{n}$$

$$\int_{-1}^3 (2x - 1) dx =$$

$$\lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \left(4 + \frac{16}{n}\right) = 4.$$

Please turn over and finish the rest of the question.

(3 points) 3. Use the Fundamental Theorem of Calculus, Part 2 to evaluate the given definite integrals:

$$(a) \int_{-1}^3 (2x - 1) dx$$

$$= [x^2 - x]_{-1}^3$$

$$= [(3)^2 - (3)] - [(-1)^2 - (-1)]$$

$$= [9 - 3] - [1 + 1]$$

$$= 6 - 2 = 4.$$

$$(b) \int_1^4 \frac{2 - \sqrt{t}}{t^2} dt = \int_1^4 \left(\frac{2}{t^2} - \frac{\sqrt{t}}{t^2} \right) dt$$

$$= \int_1^4 \frac{2}{t^2} dt - \int_1^4 \frac{t^{\frac{1}{2}}}{t^2} dt$$

$$= \int_1^4 2 \cdot t^{-2} dt - \int_1^4 t^{\frac{1}{2}-2} dt$$

$$= [-2t^{-1}]_1^4 - [-2t^{-\frac{1}{2}}]_1^4$$

$$= \left[\frac{-2}{t} \right]_1^4 - \left[\frac{-2}{\sqrt{t}} \right]_1^4$$

$$= \left[\frac{-2}{4} - \frac{-2}{1} \right] - \left[\frac{-2}{\sqrt{4}} - \frac{-2}{\sqrt{1}} \right]$$

$$= \left[-\frac{1}{2} + 2 \right] - \left[-1 + 2 \right]$$

$$= -\frac{1}{2} + 2 + 1 - 2 = \frac{1}{2}$$